

```
20.
21. Removing elements is just as straightforward:
22. # Remove by value
23. portfolio.remove("IBM") # Removes the first occurrence of "IBM"
24.
25. # Remove by index and return the value
26. sold_stock = portfolio.pop(3) # Removes and returns the item at index 3
27. print(sold_stock) # "NFLX"
28.
29. # Clear the entire list
30. portfolio.clear() # Empties the list to []
31.
32.
```

These modification operations make lists perfect for tracking dynamic information like portfolio holdings or watchlists that change as market conditions evolve.

List Comprehensions: Transforming Financial Data Efficiently

Python offers a powerful shorthand for creating new lists based on existing ones. This feature, called list comprehension, is widely used in quantitative finance for data transformation:

```
1. prices = [142.56, 143.78, 144.22, 143.86, 145.03]
2.
3. # Calculate percentage changes
4. pct_changes = [(prices[i] - prices[i-1]) / prices[i-1] * 100 for i in range(1, len(prices))]
5. # Result: [0.86, 0.31, -0.25, 0.81]
6.
7. # Identify prices above a threshold
8. high_prices = [price for price in prices if price > 144]
9. # Result: [144.22, 145.03]
10.
```

List comprehensions efficiently perform operations like calculating returns, filtering data points, or applying transformations—all common tasks in trading algorithms.